Spectral Clustering of the Irish Transmission Grid

Where the grid comes apart

Oleksandr — Team 6, Electrical Grid Hackathon 2026

Networks: `WP2024` and `WP2033`, all-island

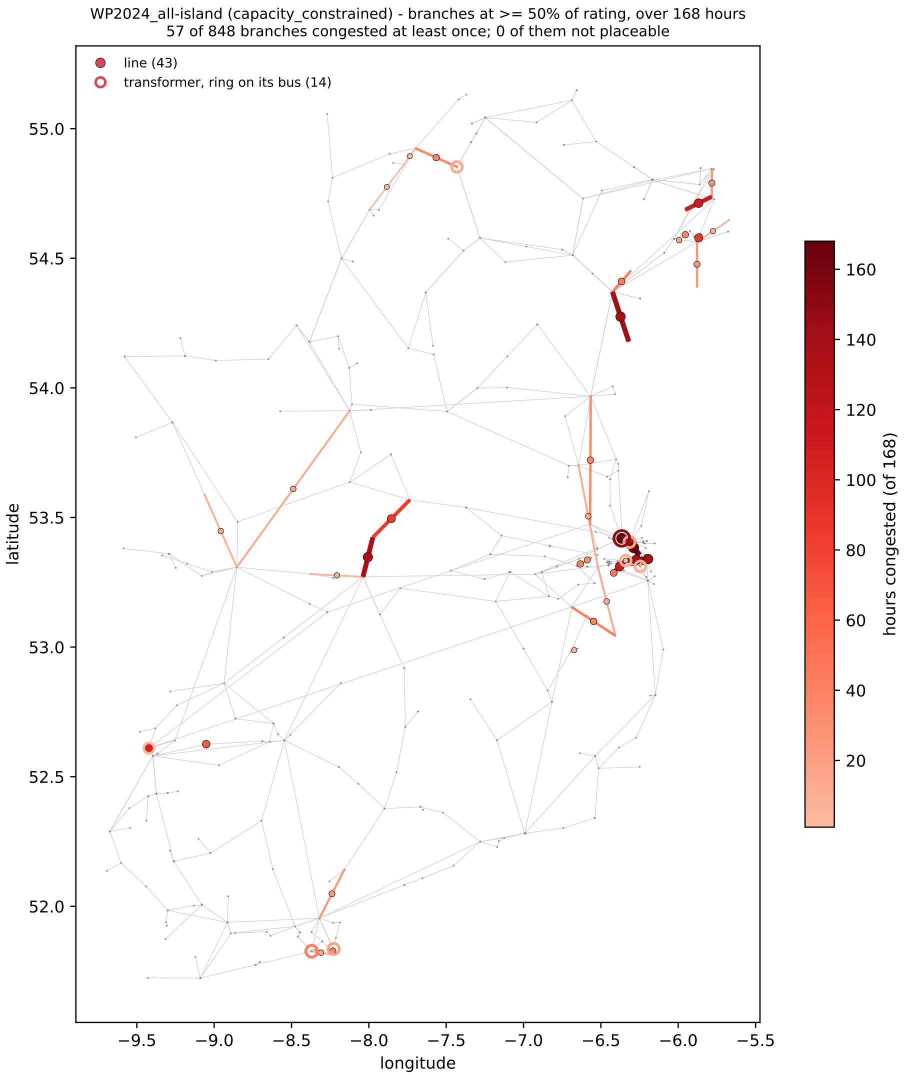
Method

1. Grid as a weighted graph: buses and generators are nodes.
2. Normalised Laplacian $L_{\text{sym}} = I - D^{-1/2} A\, D^{-1/2}$.
3. Its k smallest eigenvectors as coordinates per node.
4. k-means on those coordinates.

Edge weights

susceptance	$1/x_e$ — static network
headroom	$s_{\text{nom}} - \lvert F_e \rvert$ — one hour of dispatch

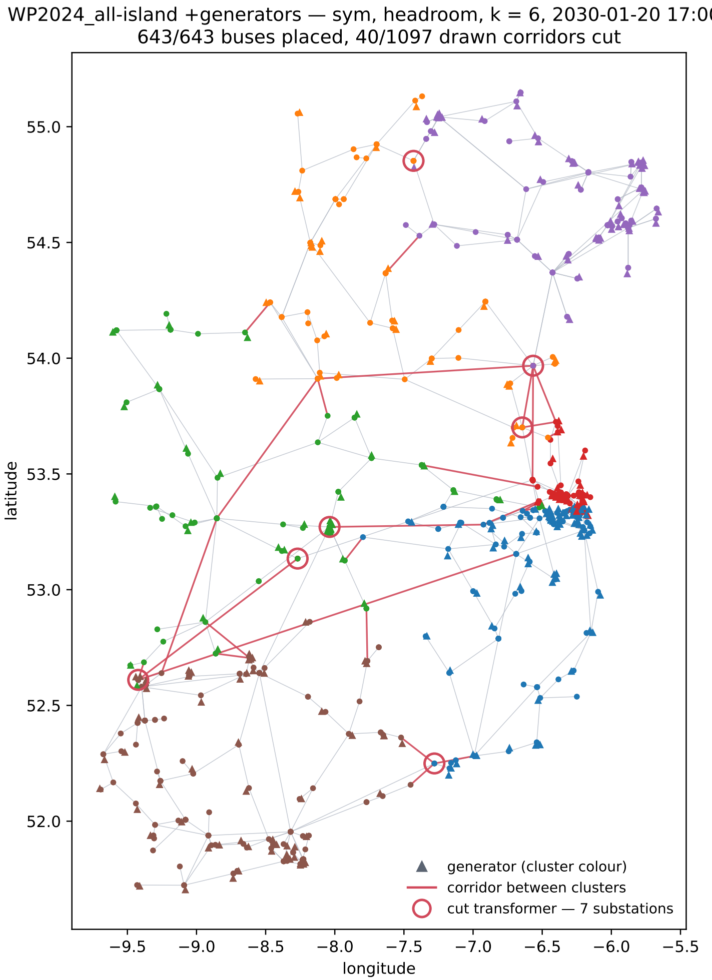
WP2024 — where the grid is loaded

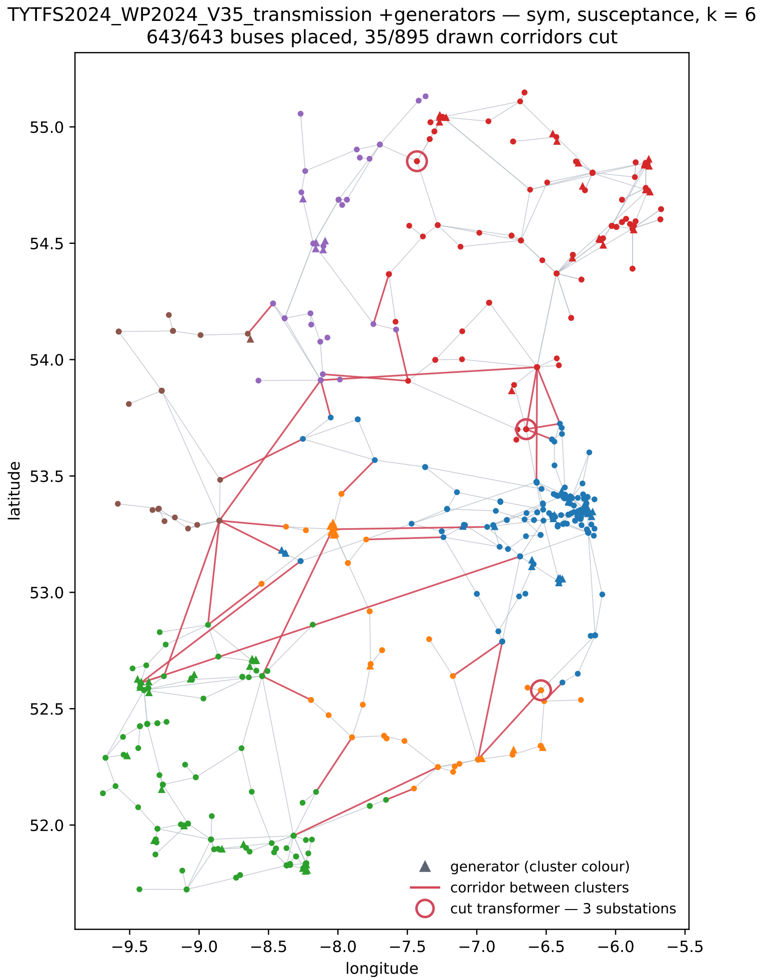


Branches at $\geq 50\%$ of rating over one week.

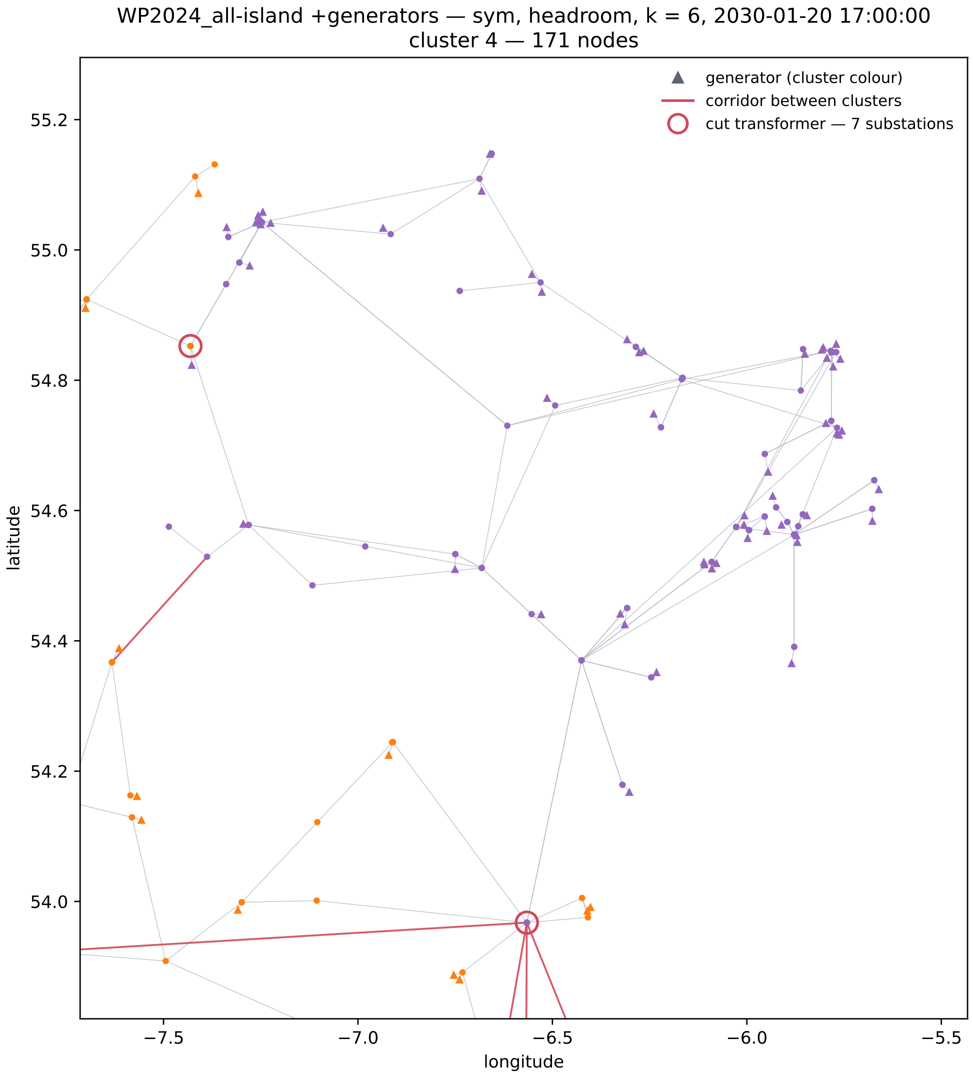
57 of 848 branches.

WP2024 — clustering, $k = 6$: headroom and susceptance



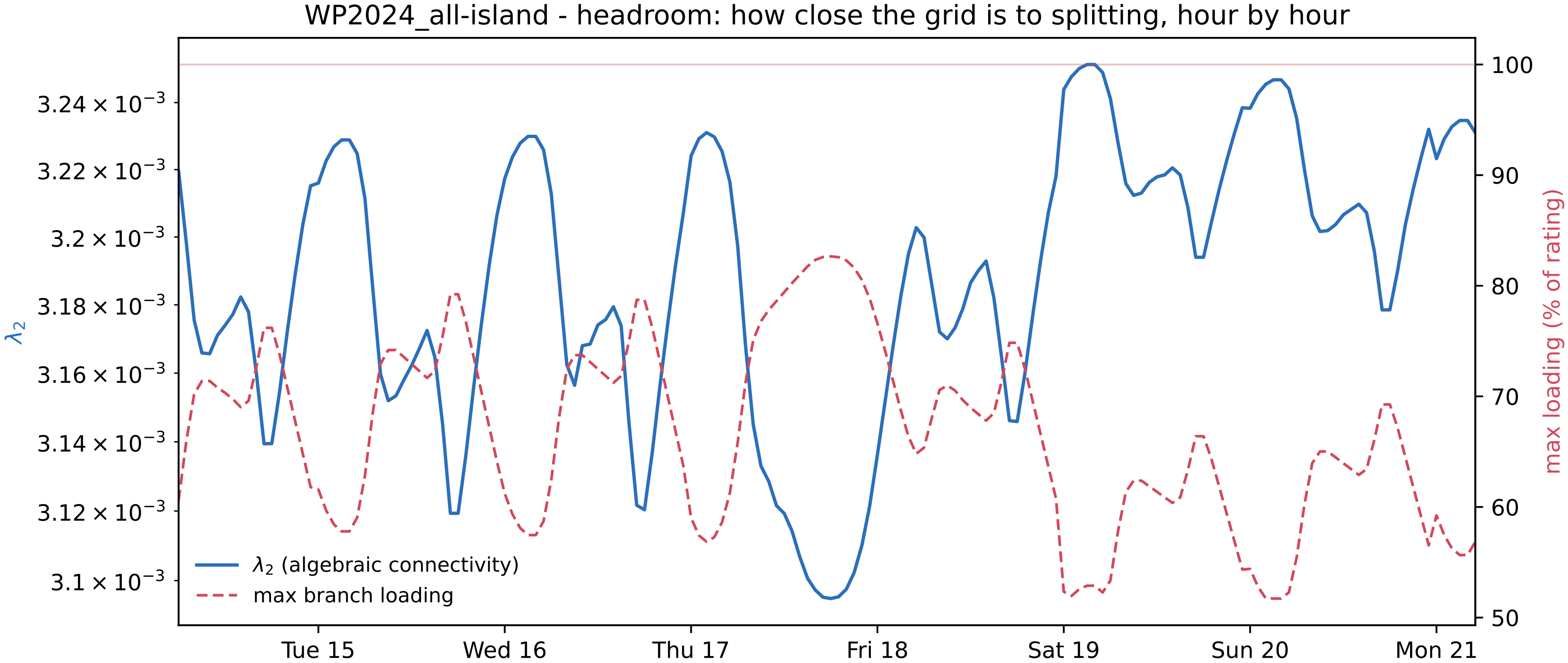


WP2024 — the north-east cluster



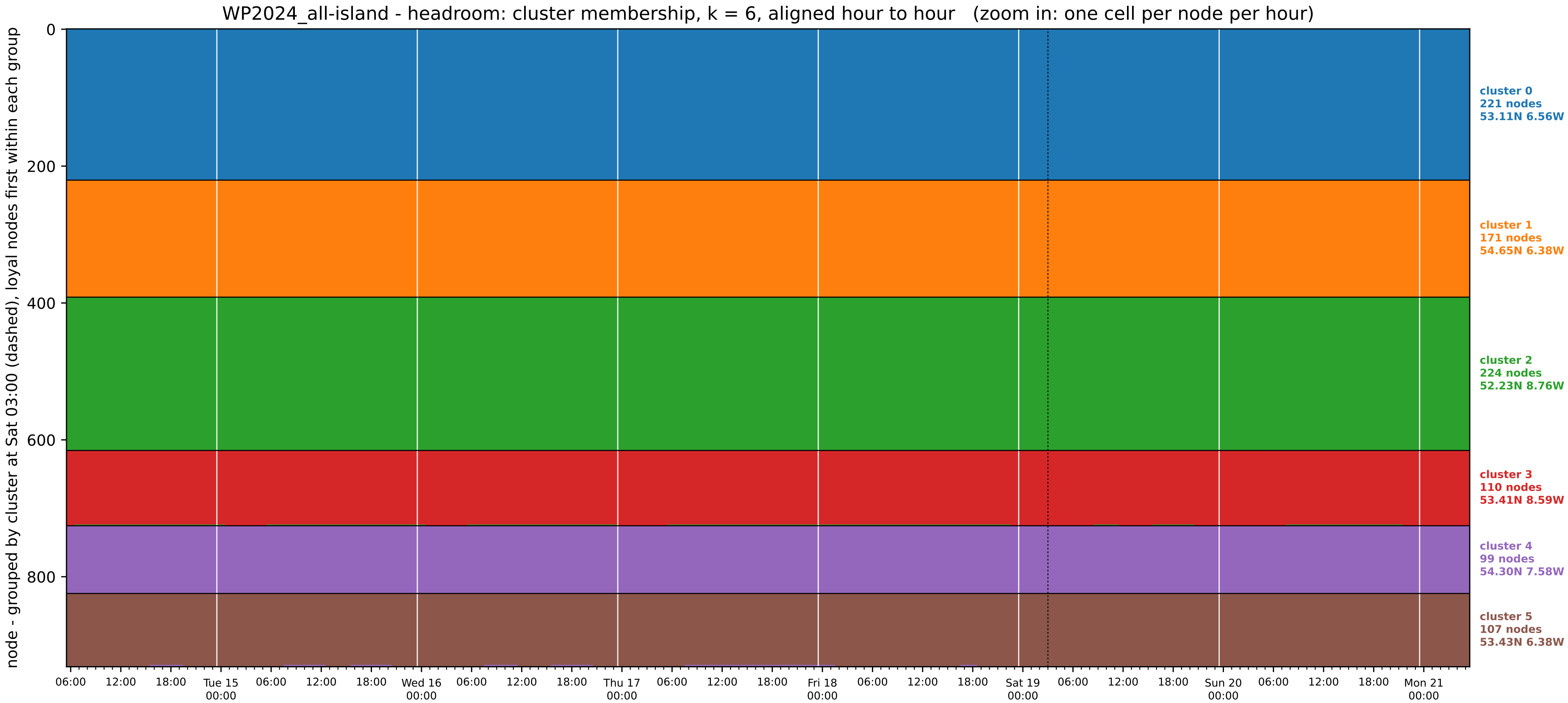
171 nodes, linked to the rest through a few corridors.

WP2024 — the week: how close the grid is to splitting



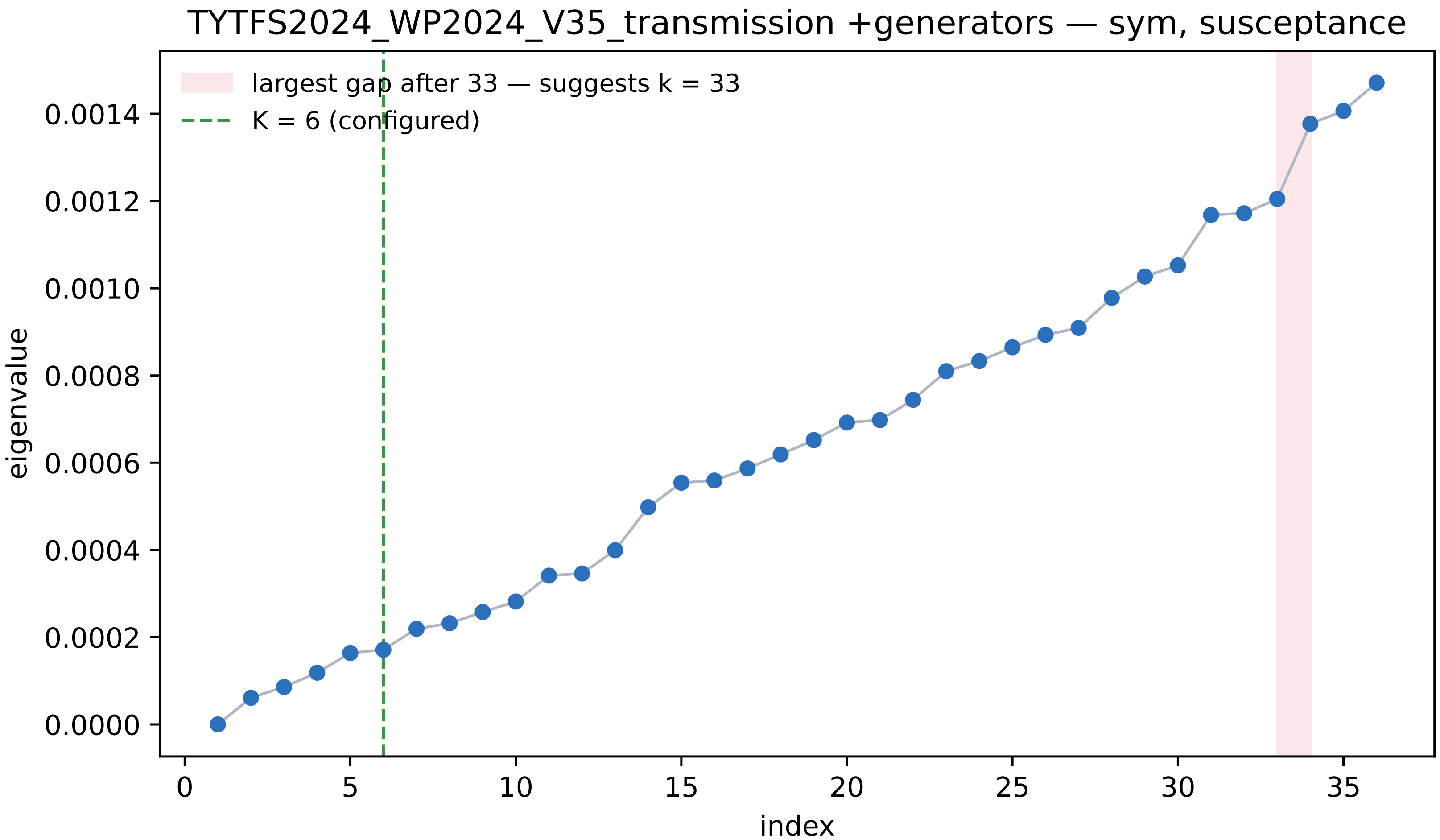
λ_2 dips whenever the maximum branch loading peaks.

WP2024 — the week: does the partition move?



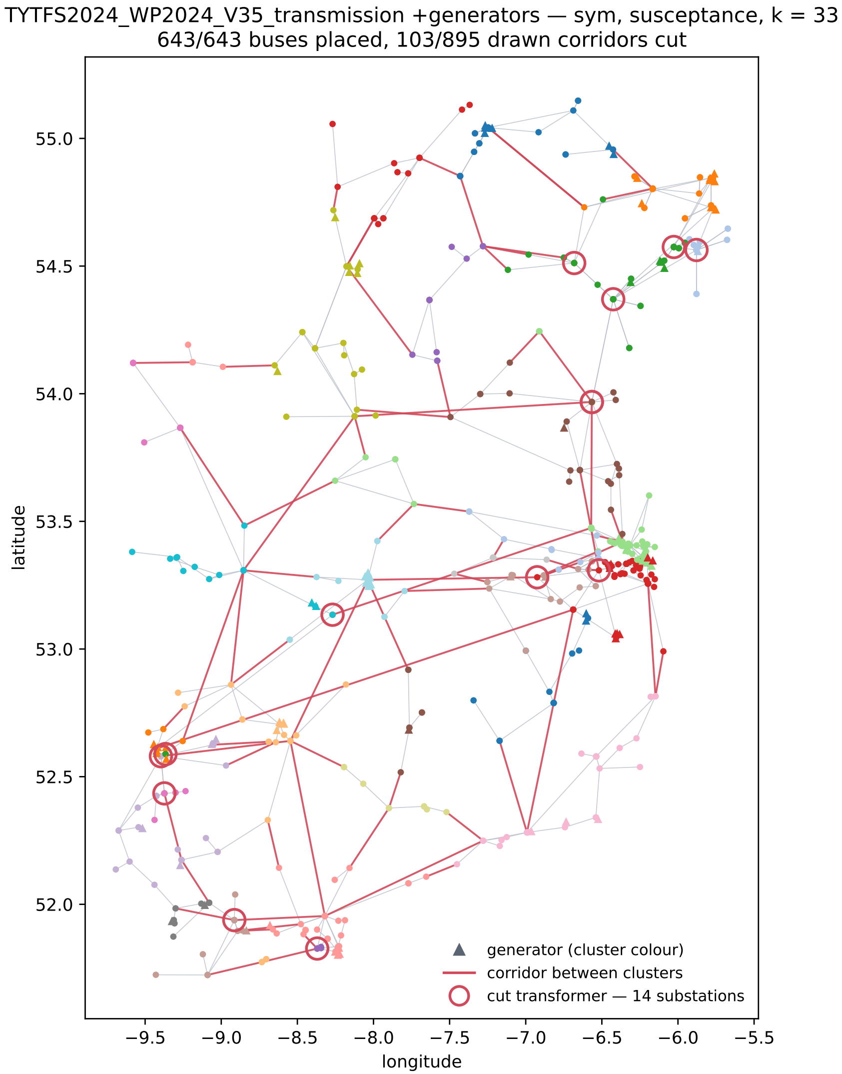
No node changes cluster over the 168 hours.

WP2024 — the eigenvalues: how many clusters?



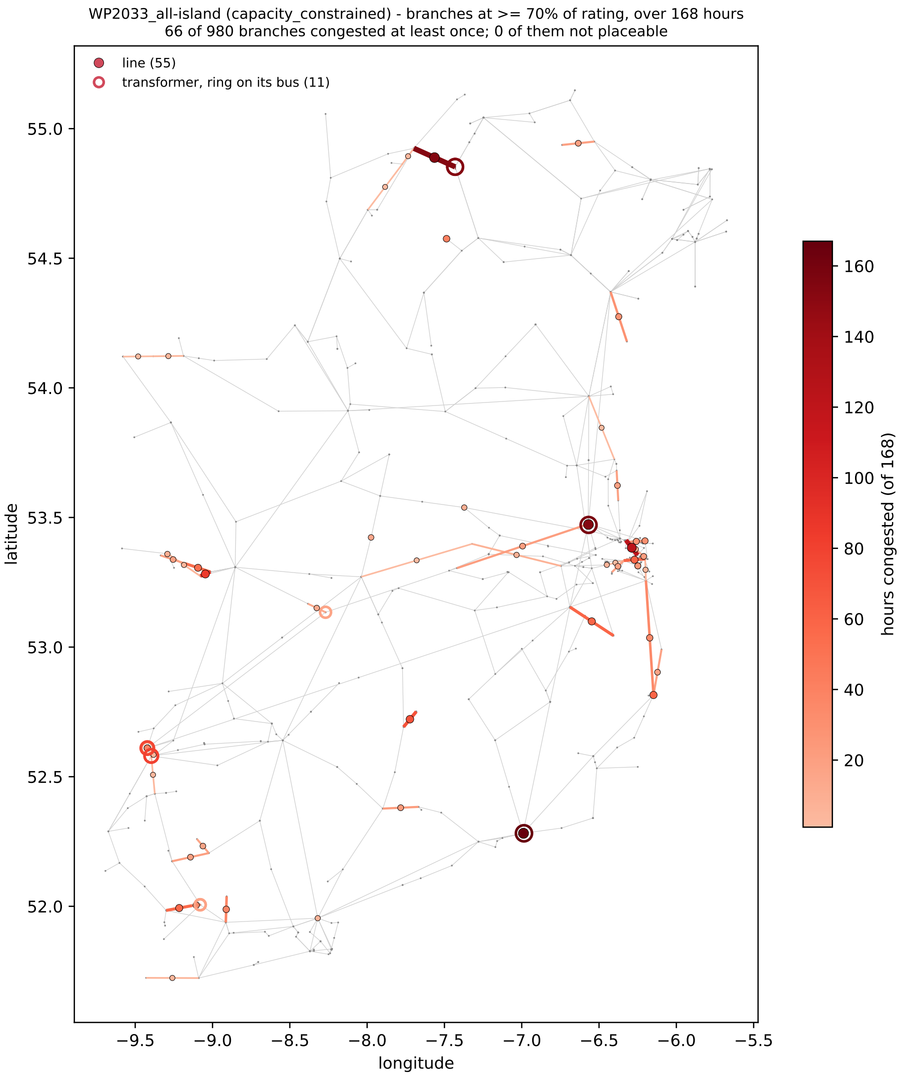
No gap at $k = 6$; the largest gap is after 33.

WP2024 — susceptance clustering, $k = 33$



k taken from the eigengap.

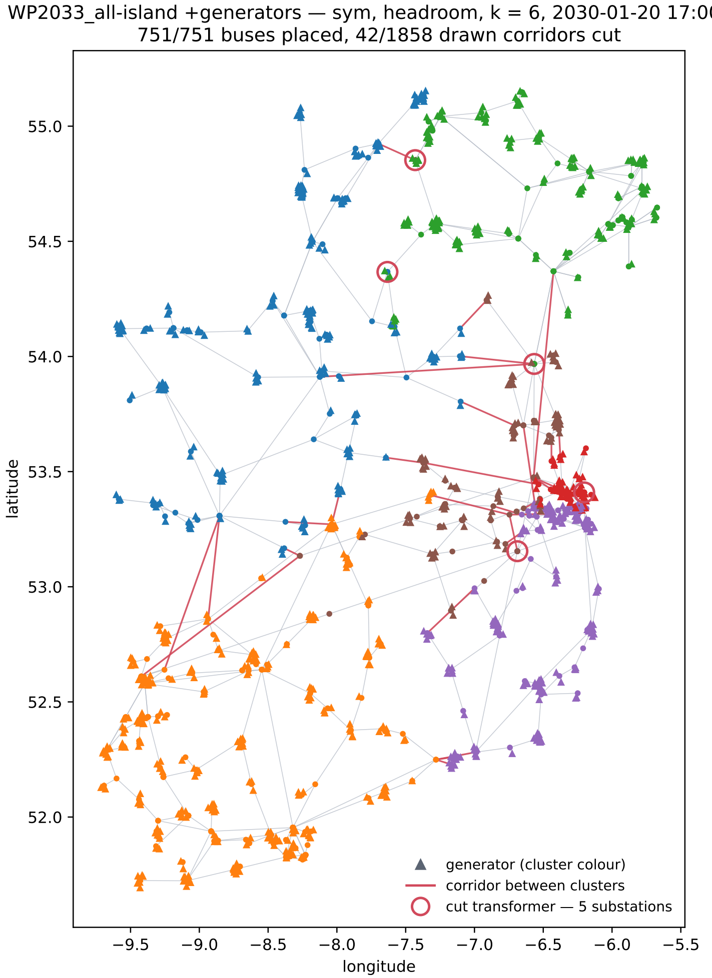
WP2033 — where the grid is loaded

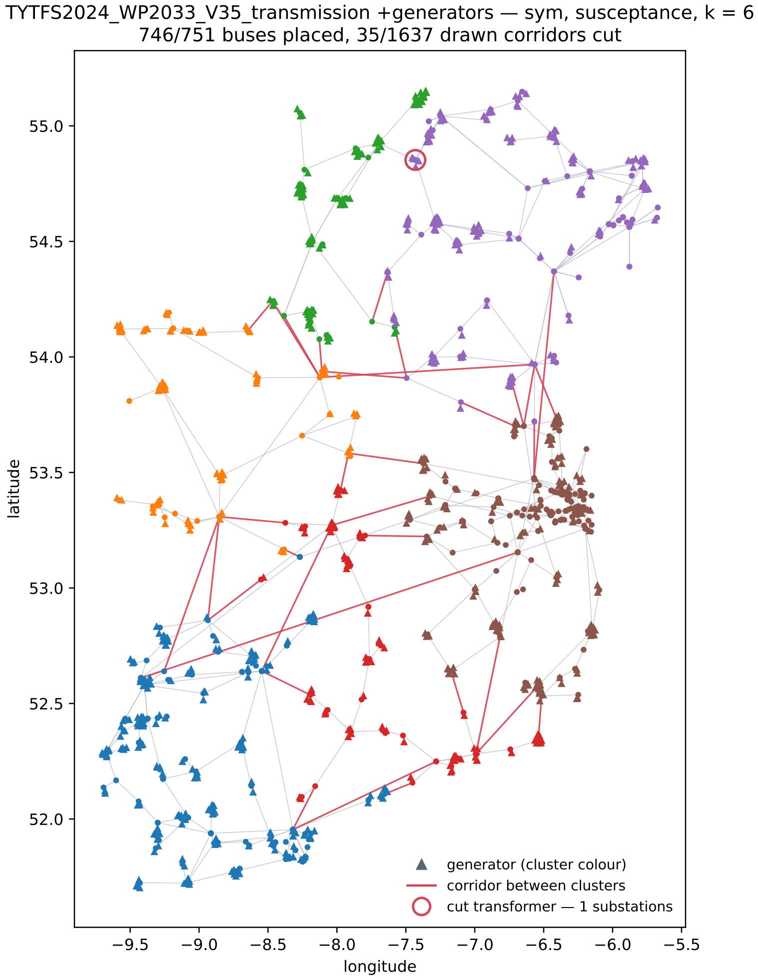


Branches at $\geq 70\%$ of rating over one week.

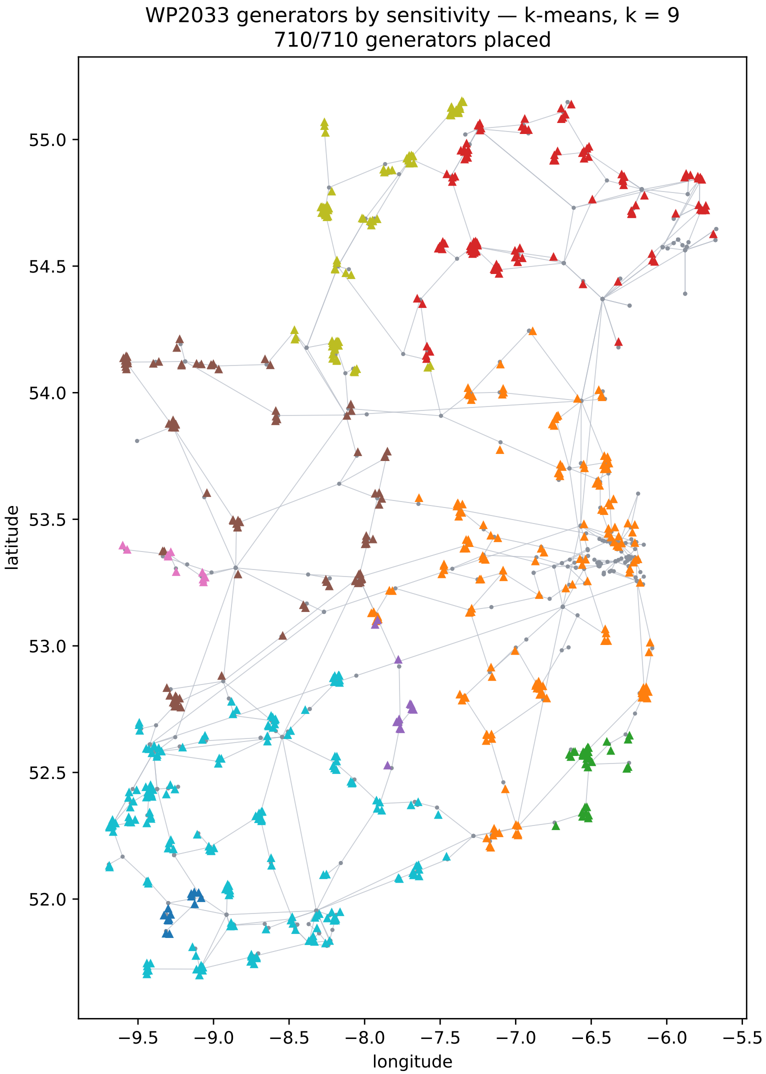
66 of 980 branches.

WP2033 — clustering, $k = 6$: headroom and susceptance





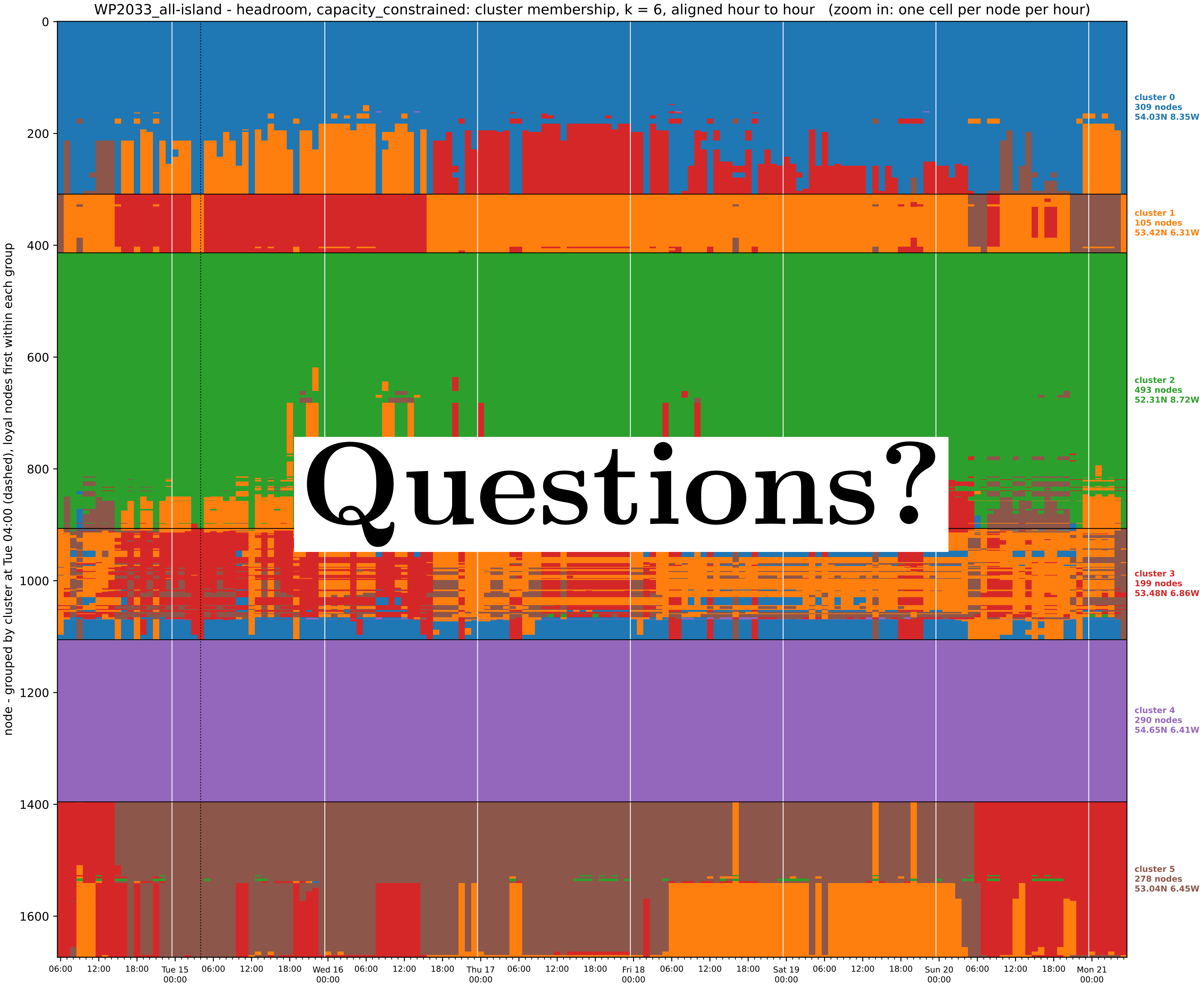
WP2033 — generators grouped by shift factor, $k = 9$



k-means on each generator's shift factors across all 980 branches.

NEXT

Questions?



WP2033_all-island - headroom, capacity_constrained: cluster membership, k = 6, aligned hour to hour (zoom in: one cell per node per hour)
node - grouped by cluster at Tue 04:00 (dashed), loyal nodes first within each group
0
200
400
600
800
1000
1200
1400
1600
06:00
12:00
18:00
Tue 15 00:00
Wed 16 00:00
Thu 17 00:00
Fri 18 00:00
Sat 19 00:00
Sun 20 00:00
Mon 21 00:00
cluster 0 309 nodes 54.03N 8.35W
cluster 1 105 nodes 53.42N 6.31W
cluster 2 493 nodes 52.31N 8.72W
cluster 3 199 nodes 53.48N 6.86W
cluster 4 290 nodes 54.65N 6.41W
cluster 5 278 nodes 53.04N 6.45W